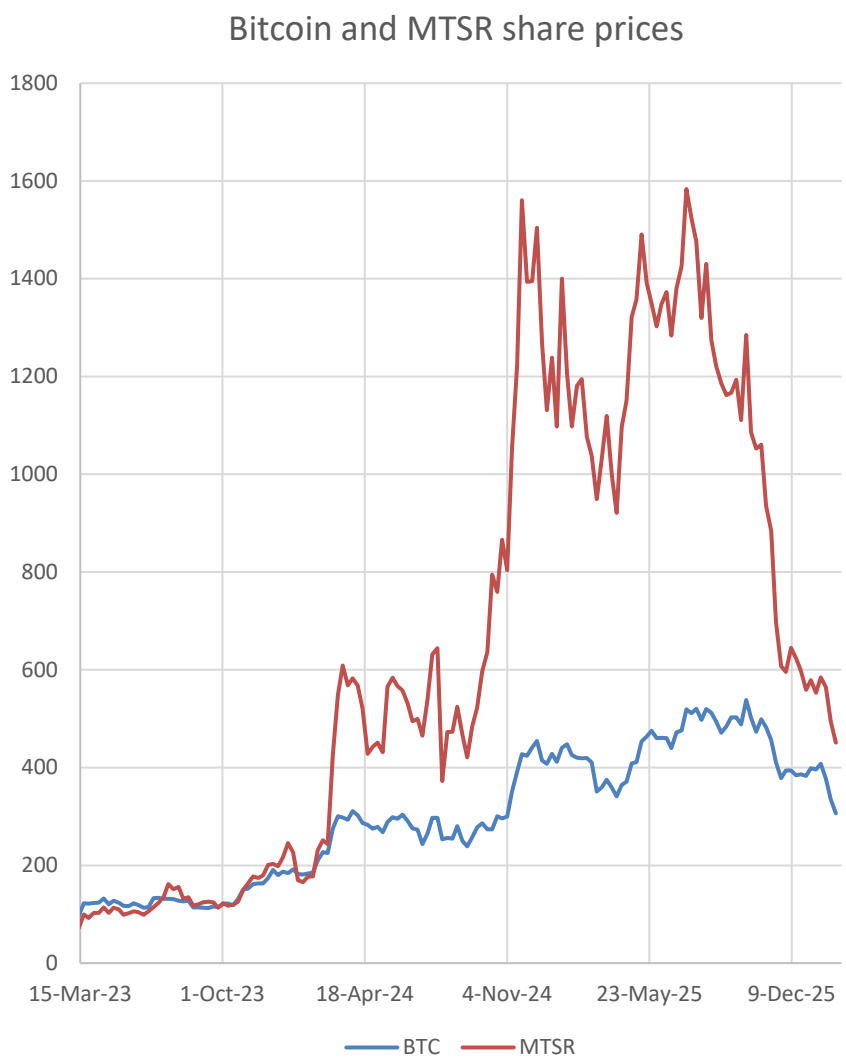
Today (last week) in Finance



- Strategy, the world's largest Bitcoin treasury company…bought even more BTC last week even as its nearly $50 billion stash remains underwater following last week's crypto market plunge.
- Strategy holds 714,644 BTC, or about 3.4% of the entire Bitcoin supply…currently valued at about $49.6 billion. The firm spent more than that to acquire the coins giving it a roughly $4.8 billion unrealized loss on its holding.
- Strategy is levered (it borrows to buy bitcoin)
- As of 8 AM today the joint slide continues

BEM 103

Class 14:
Banks as Levered investors

Key concepts

Bank balance sheets

Transformation

Bank runs (SVB)

Regulators

Banks

- Banks have multiple functions
  - payments at a distance
  - lending
- Banks make loans
  - So they need sources of funding
  - Cheapest is demand deposits from customers
  - Next, term deposits and bonds
  - Next, new equity
- Problems that banks encounter (risk)
  - Bad loans
  - Unexpected withdrawals
  - Reserve capital losses.

Balance sheet

Financial firm

- Assets
  - Loans
  - Bonds
  - Stocks
  - Derivatives
- Liabilities
  - Deposits
  - Loans
  - Equity
- Balance sheet requires A+L=0
  - Because E=A-D-L

Pre 1999 commercial bank

- Assets
  - Loans (90%)
    - Consumer
    - Business
    - Mortgages
  - Reserve funds (10%)
    - Short term bonds
    - Long term bonds
- Liabilities
  - Demand Deposits
  - Term Deposits
  - Equity
- E=0 => bank insolvent

Transformation

- Bank spread
  - Liabilities deposits earn 0.25%, CDs 4.25%
  - Assets: Auto loans 5.69%, Mortgages 6%, Credit cards 20%.
- All loans are term
  - 3 months to 30 years
- Make more money with short-term deposits and longer-term (and riskier loans).
  - Use short-term funds (deposit) to make long-term loans.
  - Bank balance sheets have a maturity mismatch.
  - So may be insolvent if deposits withdrawn too rapidly

Transformation 2

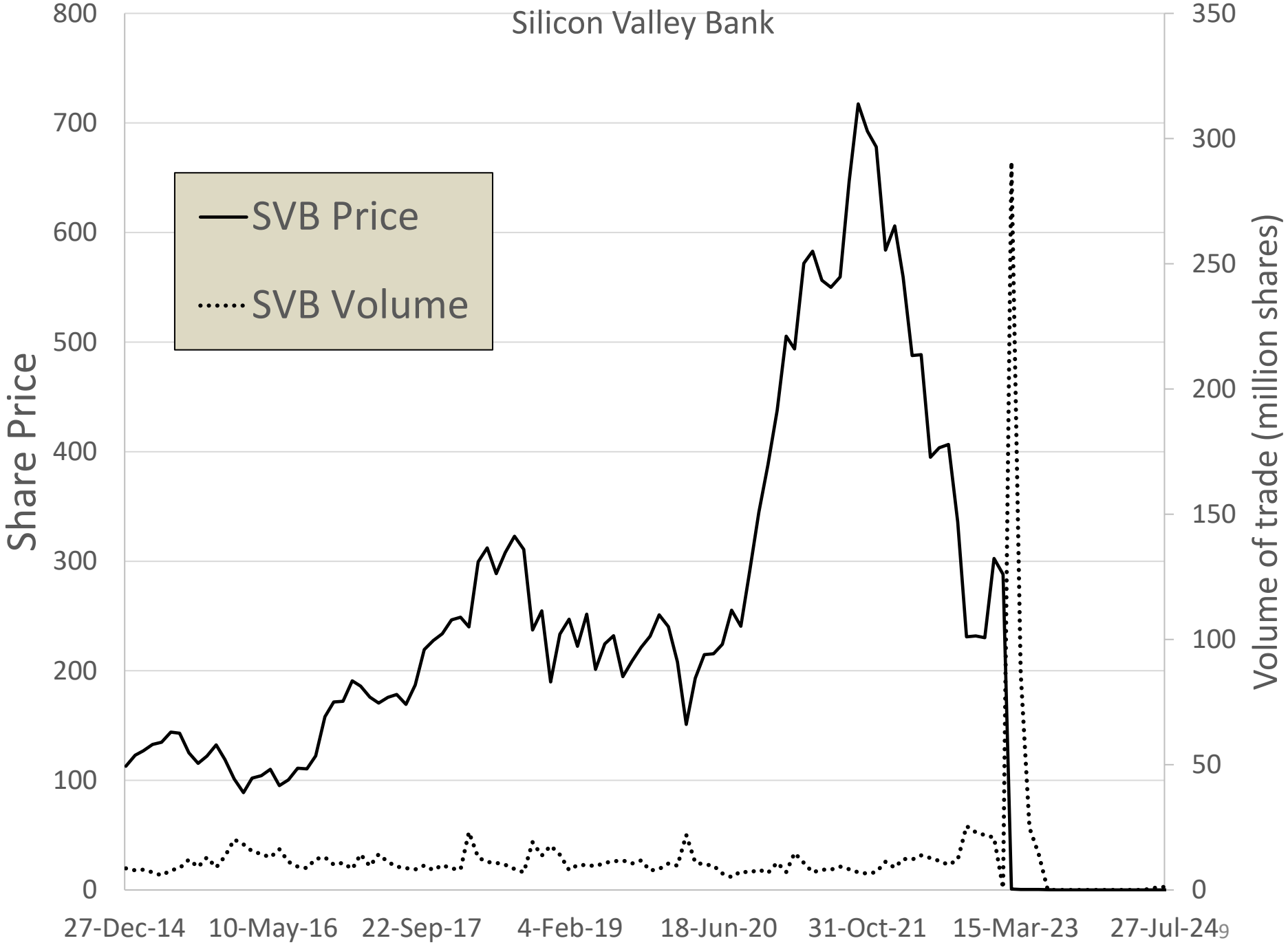
- profitable $E(r) > r_f$
- And risky
  - Investment risk
  - Liquidity risk
  - The two are related, but technically different
- Problem 1: unanticipated withdrawals
  - When someone withdraws their deposits banl must liquidate some assets
  - But many of assets are not liquid (large bid ask spread)
  - Bank should have a reserve fund in liquid assets that should be related to the distribution of net flows (new deposits –withdrawals)

Transformation 3

- Problem 2: Bad loans
  - Some borrowers do not pay
    - The Container store
    - Individual circumstance (health shocks)
    - The Great Recession in real estate
    - Shocks to economic sectors
  - Equity declines (because losses reduce equity)
  - Losses get large enough equity goes to zero
    - Then depositors start loosing money

Bank runs

- Depositors face a dilemma
  - As long as bank is healthy, keep money where it is because its (time) costly to move accounts.
  - If bank might have problems, and enough people withdraw their money, the bank becomes insolvent, accounts frozen
  - There is a rumor of problems where it pays to withdraw *even* if the bank is fundamentally healthy
- Asymmetric risk leads to bank runs
  - If I see others withdrawing their money, I want to join the queue,
  - It used to be had to go to the bank
  - Now can do this electronically.

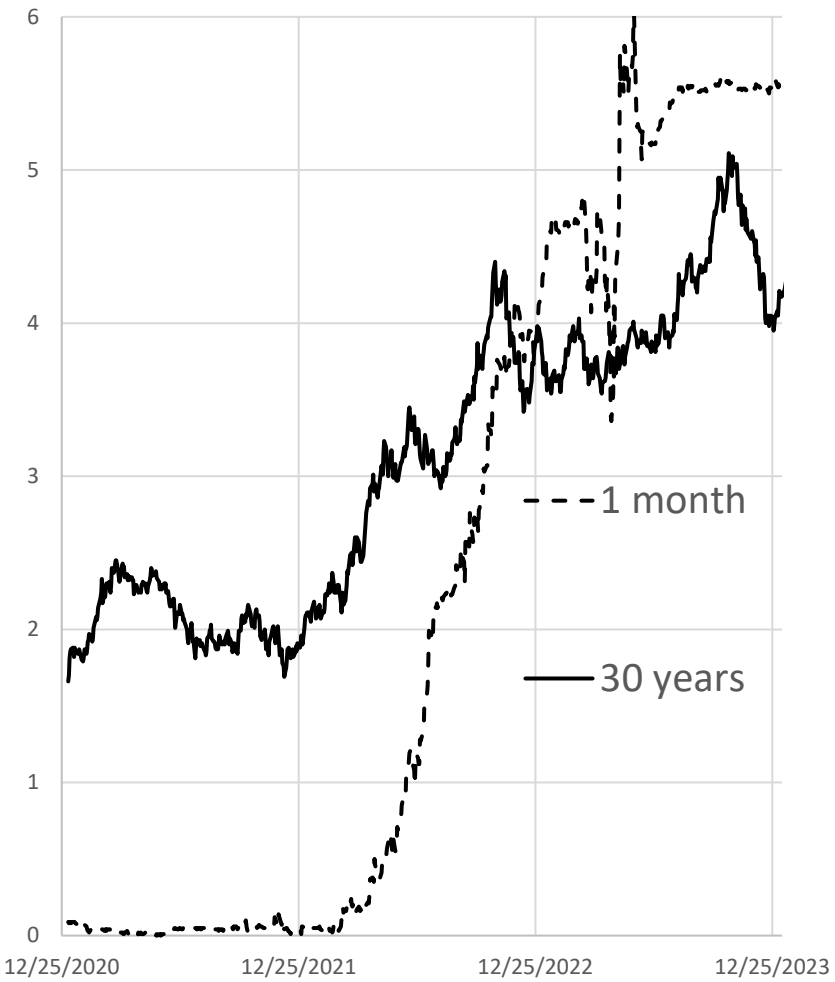

Silicon Valley Bank
SVB Price
SVB Volume
Share Price
Volume of trade (million shares)
800
700
600
500
400
300
200
100
0
350
300
250
200
150
100
50
0
27-Dec-14
10-May-16
22-Sep-17
4-Feb-19
18-Jun-20
31-Oct-21
15-Mar-23
27-Jul-24

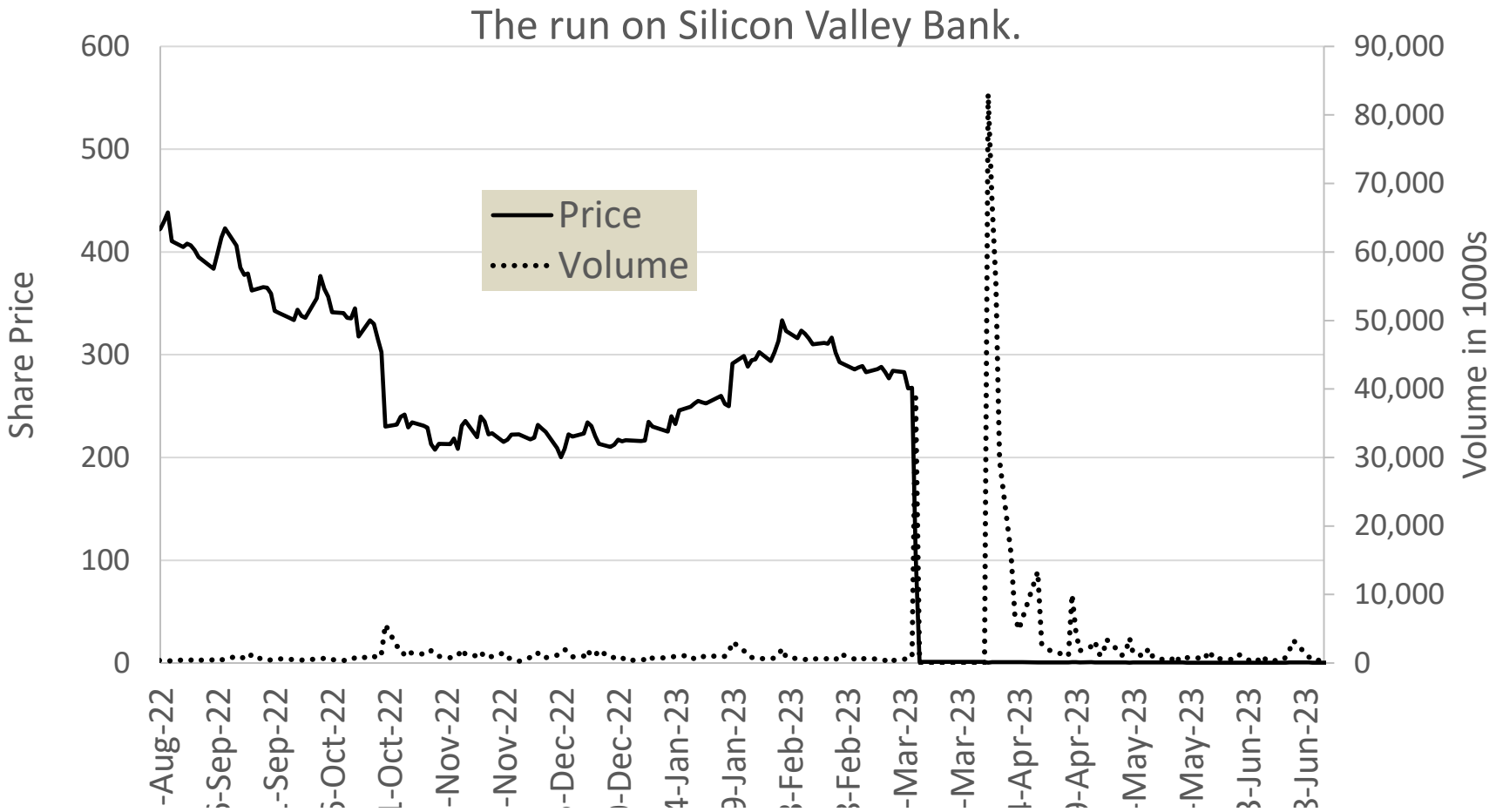
Silicon Valley Bank

- Bank founded to service start-ups on both sides of balance sheet
  - Major liabilities are deposits of Start-ups VC funds
  - 2021 $189 Billion in deposits and 125.9 in demand
  - 2022 $173 Billion in deposits and 80.7 in demand
  - Major assets are loans to Start-ups
  - Reserves held in short-term government securities
- Two problems
  - Return on reserves is very low
  - Spread between demand deposits and loans in very large

Silicon Valley Bank

- Maximizing the spread
  - Move reserves to 30 US bonds
  - Problem if interest rates rise
  - So buy put options (rights to sell) if interest rises. This is a cost.
- Alternative change accounting method for reserve
  - Mark to market vs hold to maturity
  - Feasible because SVB is not “too big to fail”
- Consequence
  - When interest rise unrealized losses
  - Depositors figure it out and withdraw 42 billion in one day




The run on Silicon Valley Bank.
Share Price
Volume in 1000s
Price
Volume
600
500
400
300
200
100
0
90,000
80,000
70,000
60,000
50,000
40,000
30,000
20,000
10,000
0
-Aug-22
-Sep-22
-Sep-22
-Oct-22
-Oct-22
-Nov-22
-Nov-22
-Dec-22
-Dec-22
-Jan-23
-Jan-23
-Feb-23
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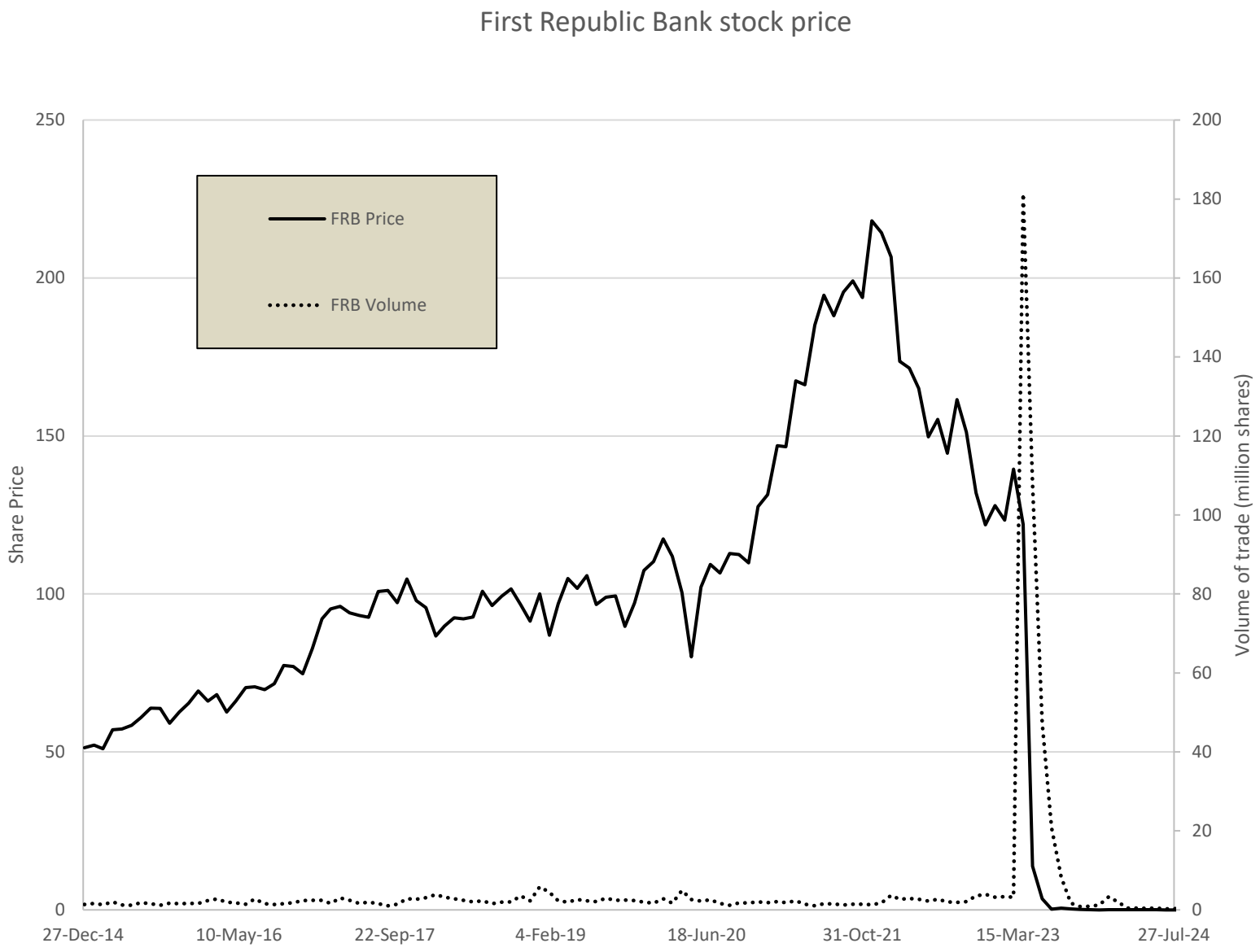
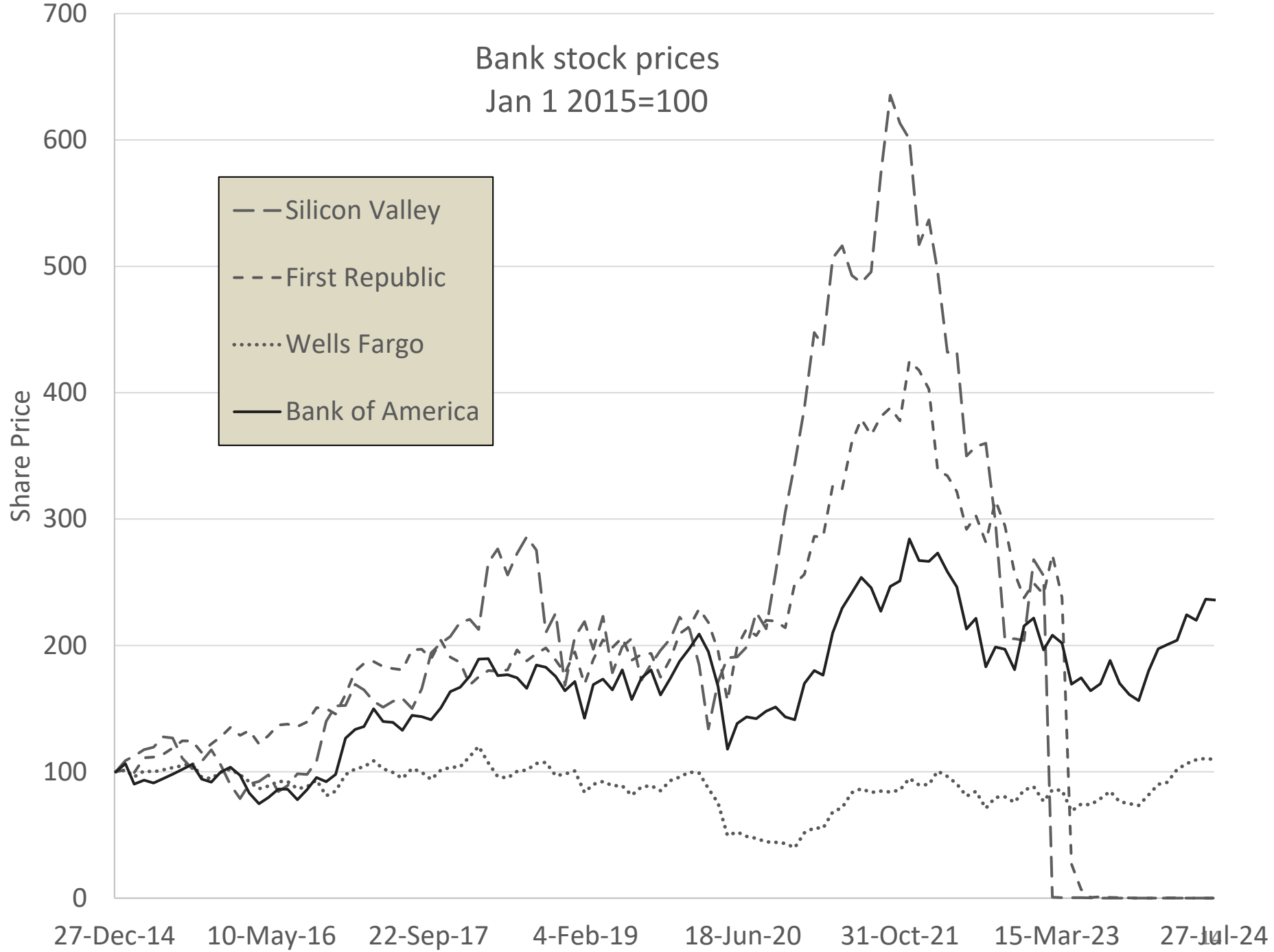
-Jun-23
-Jun-23


First Republic Bank stock price
FRB Price
FRB Volume
Share Price
Volume of trade (million shares)
250
200
150
100
50
0
200
180
160
140
120
100
80
60
40
20
0
27-Dec-14
10-May-16
22-Sep-17
4-Feb-19
18-Jun-20
31-Oct-21
15-Mar-23
27-Jul-24


Bank stock prices
Jan 1 2015=100
Silicon Valley
First Republic
Wells Fargo
Bank of America
Share Price
700
600
500
400
300
200
100
0
27-Dec-14
10-May-16
22-Sep-17
4-Feb-19
18-Jun-20
31-Oct-21
15-Mar-23
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Banks and risk

- Banks as risk neutral
  - The bank wants to maximize profits
  - It has limited liability and its shareholders are well diversified.
- Tradeoff between the number of loans the banks makes and the risk of insolvency.
- Simple example with two SoW solvent, insolvent
  - $\pi(n) = p(n)nX + (1-p)(0)$
  - Where n is the number of loans, X is the return to a loan if the bank is solvent and $p(n)$ is the probability the bank is solvent
  - $\frac{\partial p}{\partial n} < 0$; More loans reduce the likelihood of solvency.
  - $\frac{\partial \pi}{\partial n} = 0 = \frac{\partial p}{\partial n} nX + p(n)X \Leftrightarrow p(n)$=- $\frac{\partial p}{\partial n} n$
  - Interpretation bank sets the marginal increase in risk to the whole portfolio $\frac{\partial p}{\partial n} n$ equal to the likelihood the bank is solvent. In this example the bank does not take on more risk if X is high or low.

Banks and society

- Bank sets the marginal increase in risk to the whole portfolio $\frac{\partial p}{\partial n} n$ equal to the likelihood the bank is solvent.
- Something is missing: the cost to the depositors if the bank fails. It matters to them but not to the bankers.
- If the bank is big enough, this could cause additional problems
  - Bank runs (people worry about other banks)
  - Clearing problem (the failed bank's obligations to other banks do not clear)
  - Reduction in aggregate lending (economic downturn)
- Society is risk averse!

US financial regulation

- Comptroller of the currency
  - 1863 National currency act
  - Supervises National Banks
- Federal Reserve
  - 1913 Federal reserve act
  - Also supervises national banks and offers short term credit and deposit facilities
- FDIC
  - 1933 banking act
  - Provides insurance on deposits up to a given amount for all member banks
- State agencies
  - State banks
  - Insurance companies
  - Other financial actors

The Insurance Dilemma

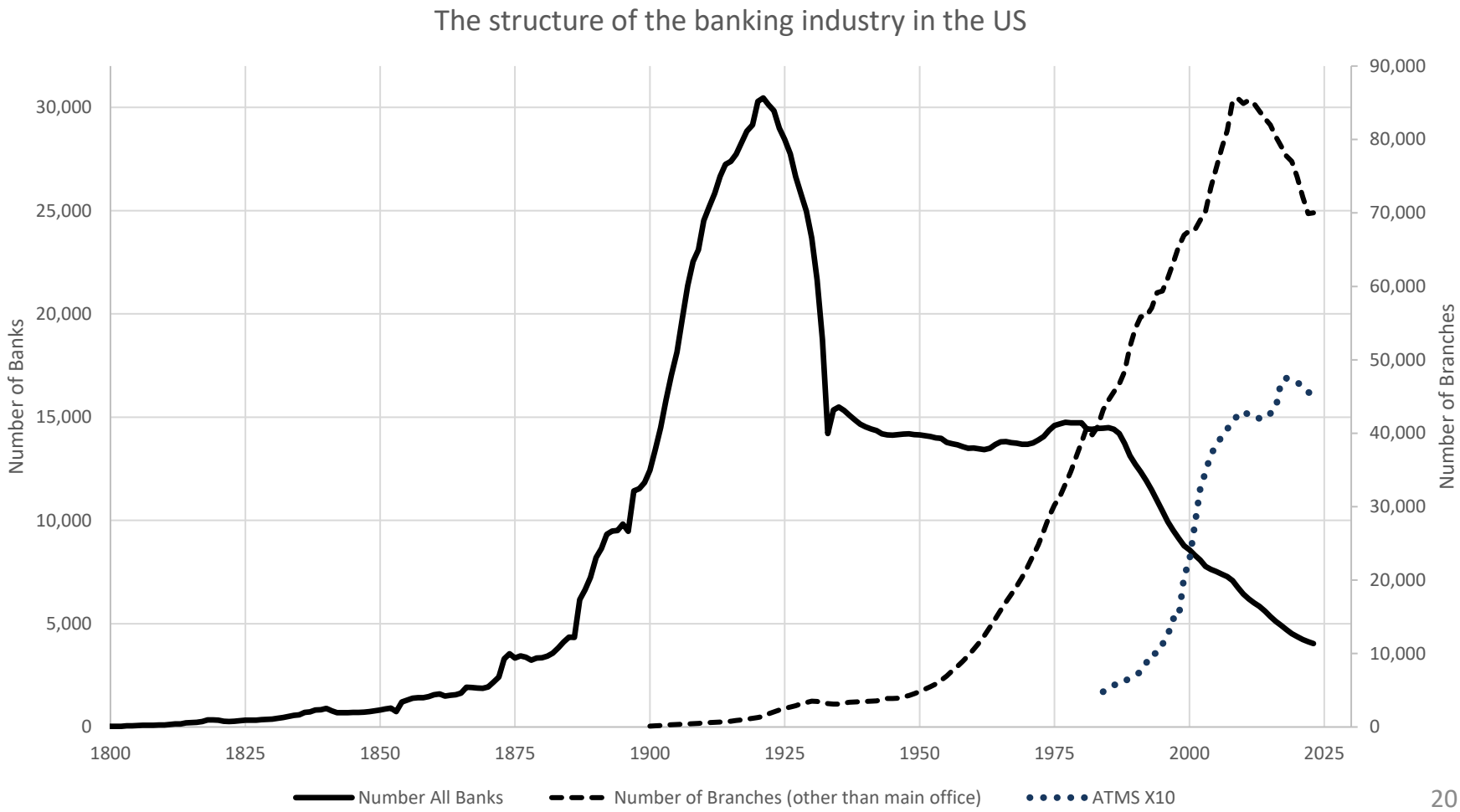
- Solution to prevent bank runs
  - FDIC insures all account up to a value ($250,000) today
  - Problem for SVB and FRB, their accounts were in the millions
  - After the failures, the FDIC insured all accounts
  - If the state does not monitor, then the customers must do it
  - If the state monitors, then customers do not
- Given insurance then monitor!
  - Balance sheet (mark to market)
  - Stress tests
  - But pressure to let small banks do their thing…

SVB and FRB vs BoA and WFB

- SVB and FRB are big (210 Billion) and (212)
- They were allowed to fail, and are now part of other banks (Citizen and JP Morgan)
- Wells Fargo ($1.7 trillion) and Bank of America ($2.57 t) are part of the to big to fail group.
- The government and the federal reserve have committed to rescuing them if problems arise so they must be monitored and their room to maneuver is less
  - They have to mark to market.
  - They have to undergo stress test (to avoid FRB's pb)

Regulation and Innovation

- Top 10 banks in U.S. hold 70.5% of deposits and 71.4 % of assets
- This was not always the case



Wrap up

- Transformation and spreads
- Transformation and risk
- Entity risk
  - Asset
  - firm
- Systemic risk
  - Bank runs
  - Credit freezes

Class 15 The federal reserve

- Central banks
  - Bank regulator
    - Stress tests
    - Balance sheet regulation
    - Reserve requirements
  - Market regulator
    - Interest rate policy
    - Quantitative tools (balance sheet of the Fed)
  - Liquidity provider
    - Overnight window